

**Assignment 4:**  
**Creating Reports and Dashboards for**  
**Predictive and Prescriptive Analysis**

Group #7

Team members:

Alfonso Mojica, Cristian Camilo NF1007669

Amparo Garcia, Brenda Carolina NF1004311

Baeny Vargas, Teddy Fabrizio NF1005996

University of Niagara Falls Canada

Data Warehouse / Visualization

Instructor: Omid Isfahanialamdari

March 25, 2025

## Overview of the dataset

### Bike Sharing Dataset

The following dataset consists of categorical and numerical data to describe a bike rental business where there are days, weather conditions and the number of rentals in (casual, registered and total) per day.

**Figure 1**

*Results of the descriptive statistics.*

|       | season | yr     | mnth   | holiday   | weekday | workingday | weathersit | \ |
|-------|--------|--------|--------|-----------|---------|------------|------------|---|
| count | 731.00 | 731.0  | 731.00 | 731.00    | 731.0   | 731.00     | 731.00     |   |
| mean  | 2.50   | 0.5    | 6.52   | 0.03      | 3.0     | 0.68       | 1.40       |   |
| std   | 1.11   | 0.5    | 3.45   | 0.17      | 2.0     | 0.47       | 0.54       |   |
| min   | 1.00   | 0.0    | 1.00   | 0.00      | 0.0     | 0.00       | 1.00       |   |
| 25%   | 2.00   | 0.0    | 4.00   | 0.00      | 1.0     | 0.00       | 1.00       |   |
| 50%   | 3.00   | 1.0    | 7.00   | 0.00      | 3.0     | 1.00       | 1.00       |   |
| 75%   | 3.00   | 1.0    | 10.00  | 0.00      | 5.0     | 1.00       | 2.00       |   |
| max   | 4.00   | 1.0    | 12.00  | 1.00      | 6.0     | 1.00       | 3.00       |   |
|       | temp   | atemp  | hum    | windspeed | casual  | registered | cnt        |   |
| count | 731.00 | 731.00 | 731.00 | 731.00    | 731.00  | 731.00     | 731.00     |   |
| mean  | 0.50   | 0.47   | 0.63   | 0.19      | 848.18  | 3656.17    | 4504.35    |   |
| std   | 0.18   | 0.16   | 0.14   | 0.08      | 686.62  | 1560.26    | 1937.21    |   |
| min   | 0.06   | 0.08   | 0.00   | 0.02      | 2.00    | 20.00      | 22.00      |   |
| 25%   | 0.34   | 0.34   | 0.52   | 0.13      | 315.50  | 2497.00    | 3152.00    |   |
| 50%   | 0.50   | 0.49   | 0.63   | 0.18      | 713.00  | 3662.00    | 4548.00    |   |
| 75%   | 0.66   | 0.61   | 0.73   | 0.23      | 1096.00 | 4776.50    | 5956.00    |   |
| max   | 0.86   | 0.84   | 0.97   | 0.51      | 3410.00 | 6946.00    | 8714.00    |   |

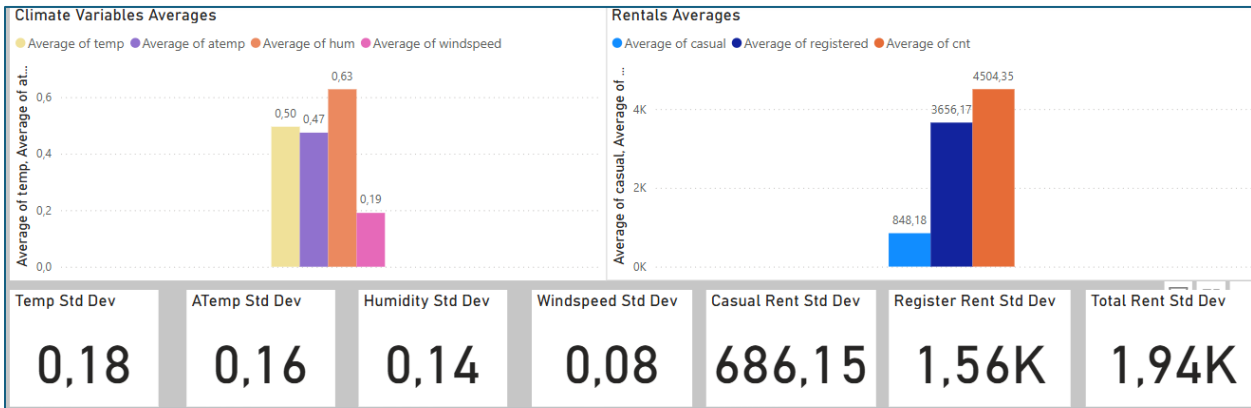
*Note.* The results of the descriptive statistics are the previous where a python command was used.

## Visualizations

With Power BI has been made graphs in to show the average and standard deviation, but only for the numerical columns which are the ones that belong to temperature, actual temperature, humidity, wind speed, casual rentals, registered rentals and total rentals (cnt).

Figure 2

*Bike Rentals & Climate Insights.*



*Note.* This dashboard provides an analytical overview of the Bike Sharing Dataset, it includes statistical summaries of temperature, humidity, and wind speed, casual and registered users.

**Key insights derived from the visualizations and advanced analysis.**

Python code has been created to look for missing values which there are none in this dataset as seen in the following graph:

Figure 3

*Look missing values in Dataset.*

```
-----
Missing values in each column:
```

```
instant      0
dteday       0
season       0
yr           0
mnth        0
holiday      0
weekday      0
workingday   0
weathersit    0
temp         0
atemp        0
hum          0
windspeed    0
casual       0
registered   0
cnt          0
dtype: int64
```

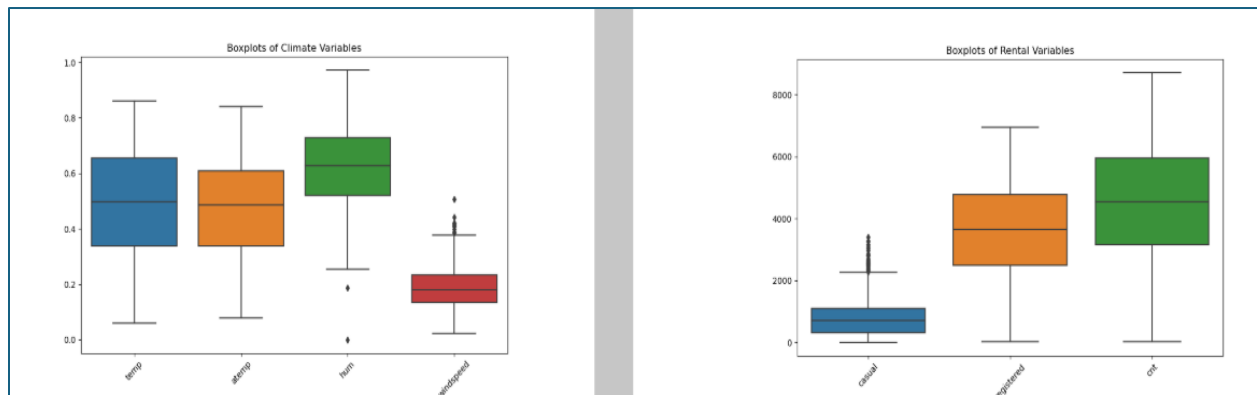
```
Total missing values: 0
-----
```

*Note.* No missing values were found.

Finally, create boxplots for the 7 numerical variables to look for outliers with the following results:

**Figure 4**

### *Boxplot Analysis*

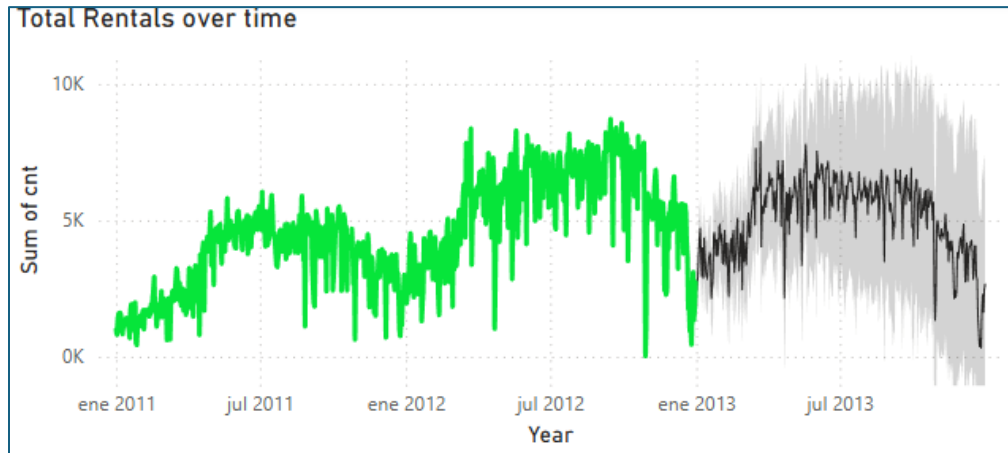


*Note.* The boxplot reveals the presence of several outliers in wind speed, the casual rental variable also shows multiple outliers, mainly on the higher end, it can be due weather conditions or some events such as holidays.

## Visualizations

**Figure 5**

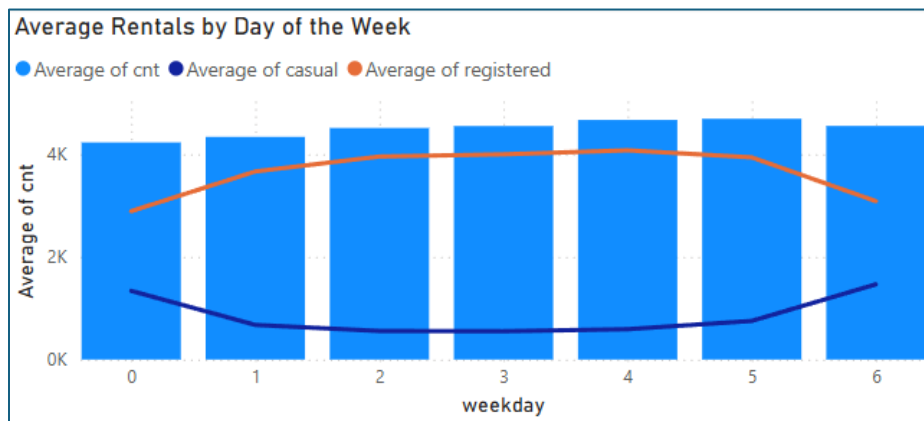
*Total Rentals over time*



*Note.* This visualization reflects the total amount of rentals over time and then make a forecast with a seasonality of 365 as it is using the daily number of rentals in the line chart. As this graph shows the sales increase during spring or summer months and decrease in winter months. The forecast reflects this feeling with the rentals following a similar pattern to the second year which is higher than the first one.

**Figure 6**

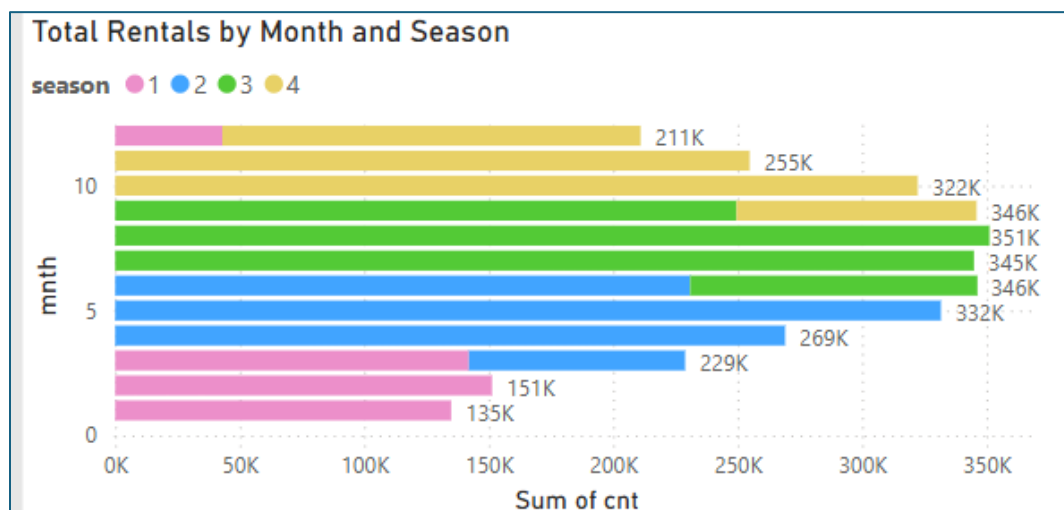
*Average Rentals by Day of the Week*



*Note.* The next graph shows the average number of rentals by day of the week. This graph is interesting because all days show similar average of rentals with Thursday (4) and Friday (5) being a little bit higher than others. Next, we also see that the registered rentals are higher during weekdays and reduce in weekends (Saturday (6) and Sunday (0)) which is the opposite for casual rentals which shows the opposite patterns.

**Figure 7**

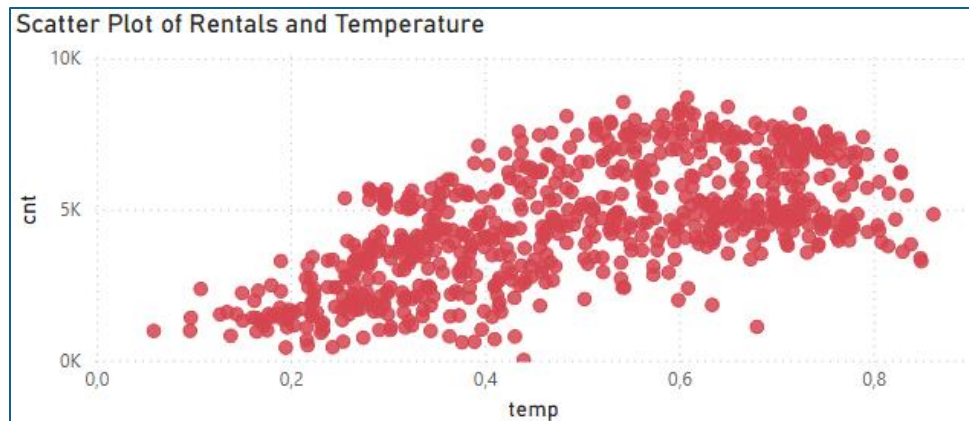
*Total Rentals by Month and Season*



*Note.* This graph visualizes the total rentals by month and season that confirms what the total rentals over timeline graph shows which is that rentals increase during spring and summer with summer being the best season for the company. Also, winter rentals are the lowest which is logical given that the season is not friendly to do this type of activities.

**Figure 8**

*Scatter Plot of Rentals and Temperature*



*Note.* The final graph here is a scatter plot between total rentals and temperature which shows that higher temperatures are linked with higher numbers of rentals which is also related to the previous insights because summer months have the highest temperatures and winter have the lowest.

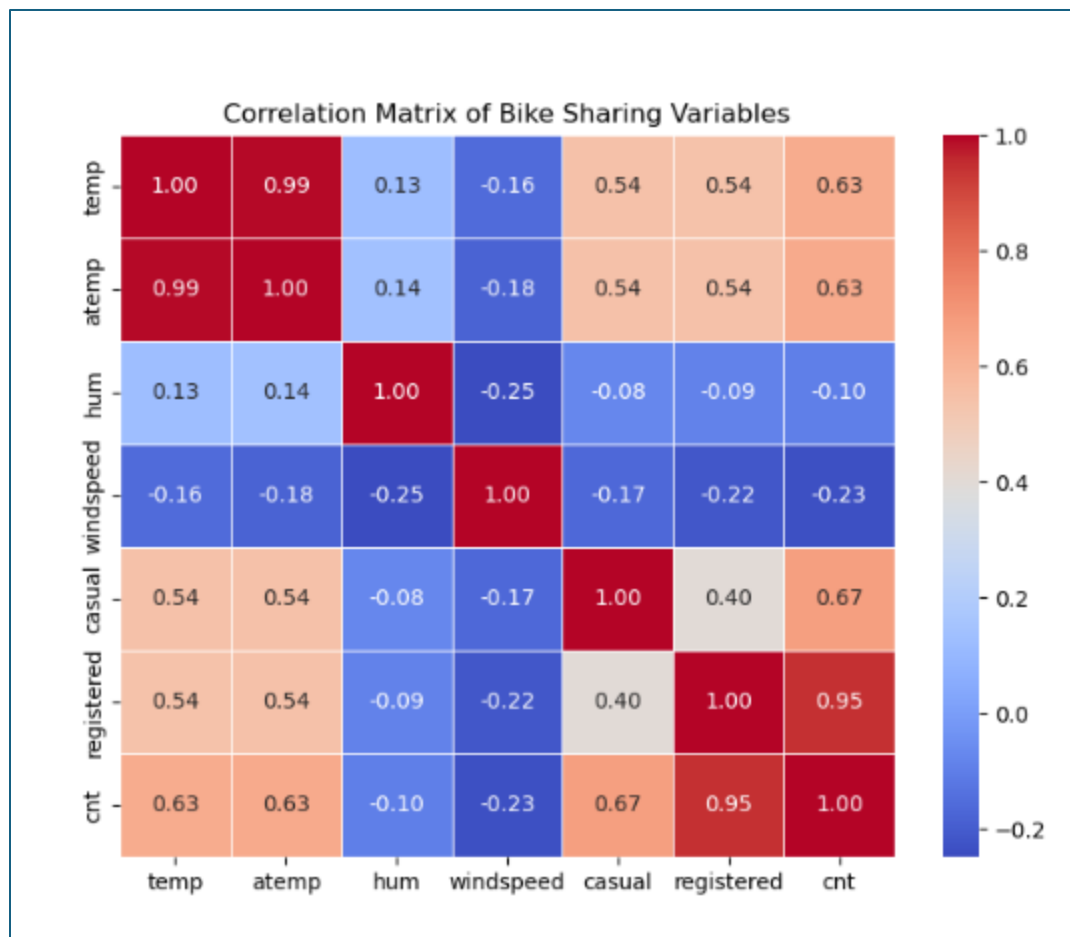
The final thing added to the dashboard was a measure to compare the yearly growth between 2011 and 2012 rentals which reflects a 64.88% growth rate which is an outstanding value.



### Key insights derived from the visualizations and advanced analysis

#### Figure 9

*Correlation Matrix for Climate and Rental Columns*

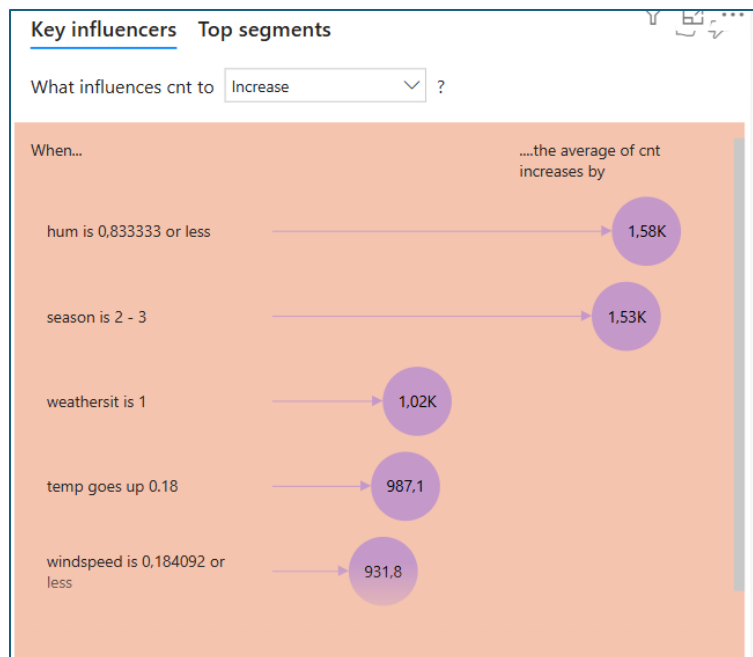


The correlation matrix shows that temperature has a high positive correlation with the rental columns (casual, registered and cnt) and humidity and windspeed have a low negative correlation value with these columns showing that increased humidity decreases rentals but not in considerable numbers.

**Figure 10**

*Key Influencers of Total Rentals*





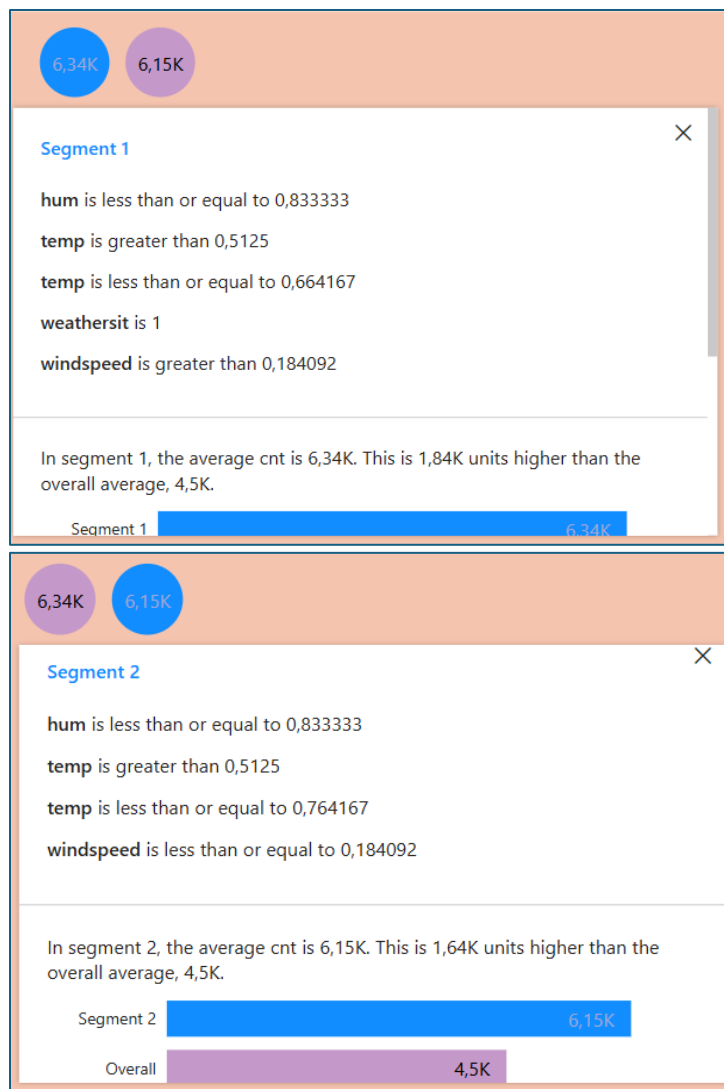
*Note.* Key influencers visualization shows which variables influence rentals and in which quantity.

The results show that when humidity is less than 83.3% rentals average increase by 1.58K a similar number when season is 2 or 3 which belong to summer and spring.

Finally, it can be defined that when weather situation, temperature and windspeed also influence in the number of rentals so considering them when making decision such as total bikes available, we need to consider all these factors into account.

## Figure 11 and 12

### *Segments for Rentals*



*Note.* In the end, we show two segments for rentals. Both are similar and show characteristics such as low or medium humidity, medium and high temperatures and medium to high wind speeds which indicate more rentals. The only difference is that weather situation is always 1 for the first segment which reflects sunny days.

### **Recommendations for further exploration or applications of the insights.**

The recommendation we can make for decision makers in this rental company is that they should increase bikes for months with good climate in spring and summer seasons. Another insight is that

weekdays don't influence heavily on rental numbers, so looking at deals for specific days is not recommended. Finally, doing a more precise forecast is suggested because the company had a huge increase in rentals from one year to the next and the forecast here shows that it will be similar to the previous year which may not be the case.

## **Overview of the dataset**

### **Housing Prices Dataset**

Is a predictive dataset used for real estate price estimation. It contains various attributes related to property characteristics, construction details, and overall quality, which influence the final sale price of a house.

The attributes have been separated into categories for easy compression as follows:

#### **1. Property Identification & Location**

- Id: Unique identifier for each property
- LotFrontage: Linear feet of street connected to the property
- LotArea: Size of the lot in square feet

#### **2. House Quality & Condition**

- OverallQual: Overall material and finish quality (1-10 scale)
- OverallCond: Overall condition of the house (1-10 scale)

#### **3. Construction & Renovation**

- YearBuilt: Year when the house was built
- YearRemodAdd: Year of the most recent remodeling

#### **4. Roof & Exterior Features**

- RoofStyle: Type of roof (e.g., Gable, Hip)
- RoofMatl: Material used for the roof
- Exterior1st, Exterior2nd: Exterior covering materials

- MasVnrType: Masonry veneer type
- MasVnrArea: Masonry veneer area in square feet

## 5. **Basement Features**

- BsmtQual: Height of the basement
- BsmtCond: General condition of the basement
- BsmtExposure: Walkout or garden level basement exposure
- BsmtFinType1, BsmtFinType2: Basement finish types
- BsmtFinSF1, BsmtFinSF2: Finished square feet in the basement
- BsmtUnfSF: Unfinished basement square feet
- TotalBsmtSF: Total basement square feet

## 6. **Heating & Utilities**

- Heating: Type of heating system
- HeatingQC: Heating quality and condition

## 7. **Living Space & Room Counts**

- 1stFlrSF, 2ndFlrSF: Square footage of first and second floors
- LowQualFinSF: Low-quality finished square feet
- GrLivArea: Above-grade living area in square feet
- TotRmsAbvGrd: Total number of rooms above ground
- BedroomAbvGr: Number of bedrooms above ground
- KitchenAbvGr: Number of kitchens above ground
- KitchenQual: Kitchen quality rating
- FullBath, HalfBath, BsmtFullBath, BsmtHalfBath: Bathroom counts

## 8. **Fireplace & Garage Details**

- Fireplaces: Number of fireplaces
- FireplaceQu: Fireplace quality
- GarageYrBlt: Year garage was built
- GarageFinish: Interior finish of the garage

- GarageCars: Number of cars the garage can hold
- GarageArea: Size of the garage in square feet
- GarageQual, GarageCond: Garage quality and condition

## 9. Outdoor & Miscellaneous Features

- PavedDrive: Paved driveway indicator
- WoodDeckSF: Square footage of the wood deck
- OpenPorchSF, EnclosedPorch, 3SsnPorch, ScreenPorch: Porch areas
- PoolArea: Square footage of the pool
- Fence: Fence type
- MiscFeature: Additional miscellaneous features
- MiscVal: Value of miscellaneous features

## 10. Sales Information

- YrSold: Year the house was sold
- SalePrice: Final selling price of the house (target variable for prediction)

```
from sklearn.linear_model import LinearRegression

# Select features that correlate with SalePrice
features = ["LotArea", "YearBuilt", "BedroomAbvGr"] # Adjust based on your dataset

# Split data into known and missing SalePrice
known = df[df["SalePrice"].notnull()]
missing = df[df["SalePrice"].isnull()]

# Train a regression model
model = LinearRegression()
model.fit(known[features], known["SalePrice"])

# Predict missing values
df.loc[df["SalePrice"].isnull(), "SalePrice"] = model.predict(missing[features])
```

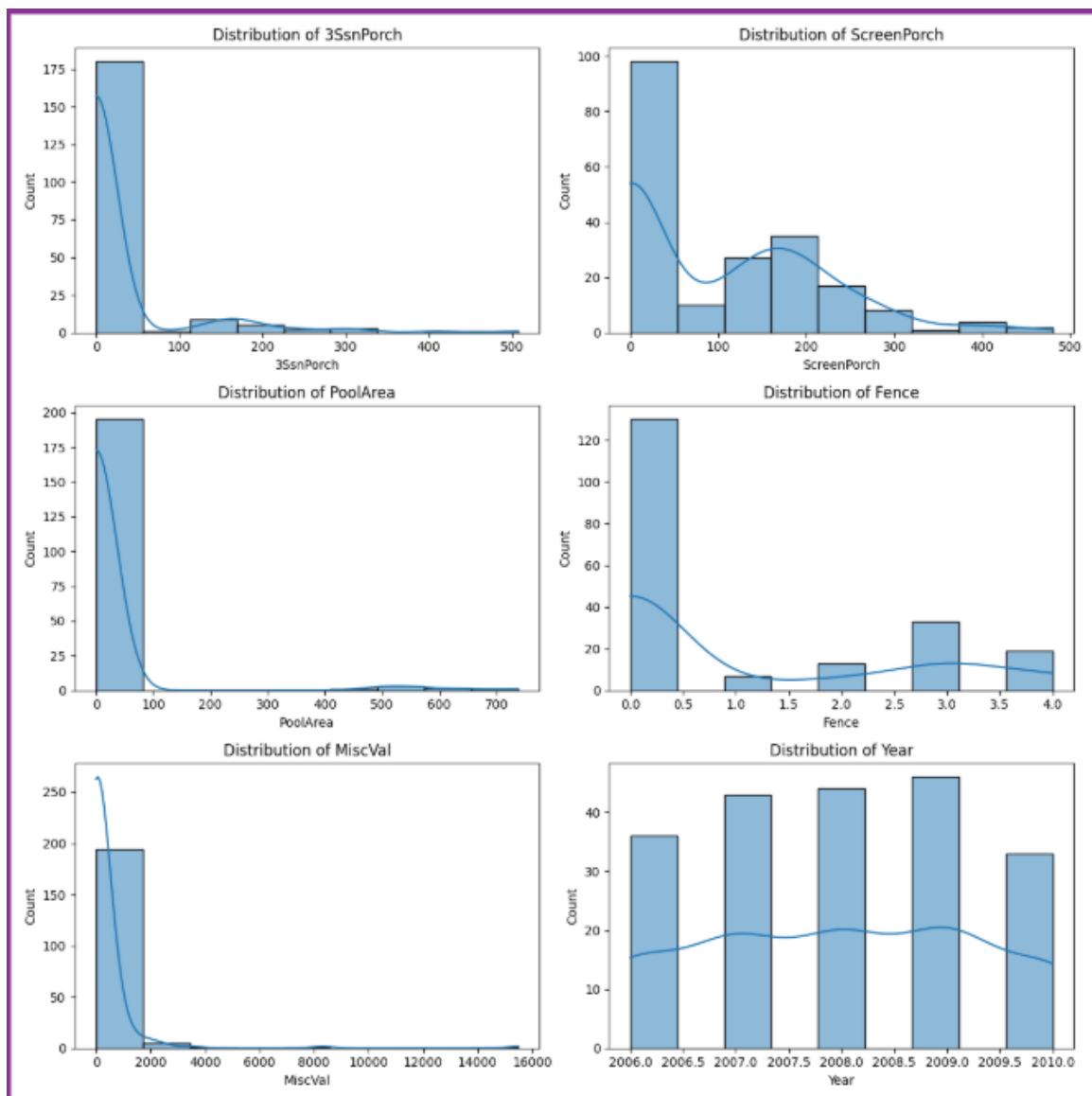
From the previous code the data was cleaned the database and enters the missing price values based on the variables used in the key insights (bubble chart).

## Visualizations

In the beginning it found missing data, so it has been created a Dax to filter all the values with “NA” after that apply a Dax function to complete all this missing data using the Mean of the Sale Price of the houses, then we run Data Statistics using the python visualization as it follows:

**Figure 13**

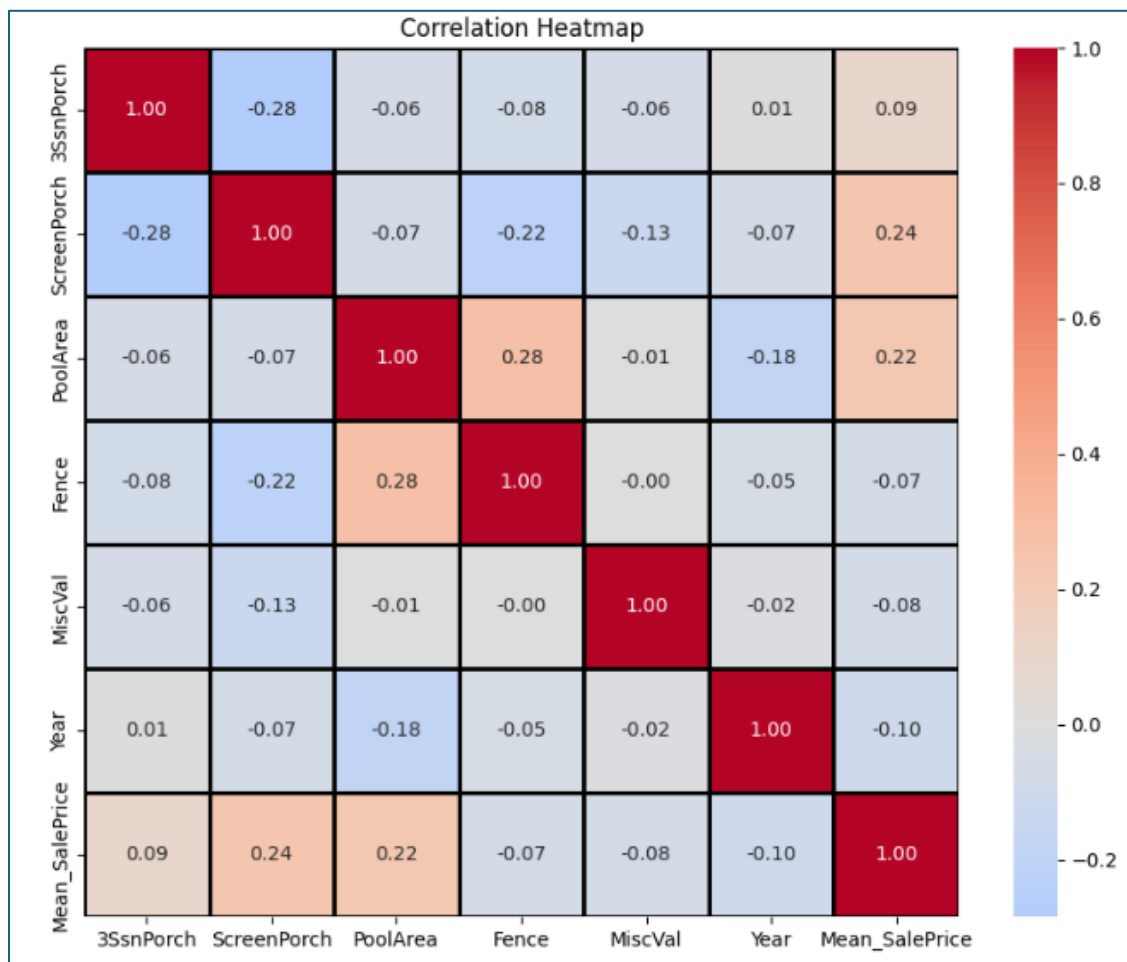
*Distribution Analysis*



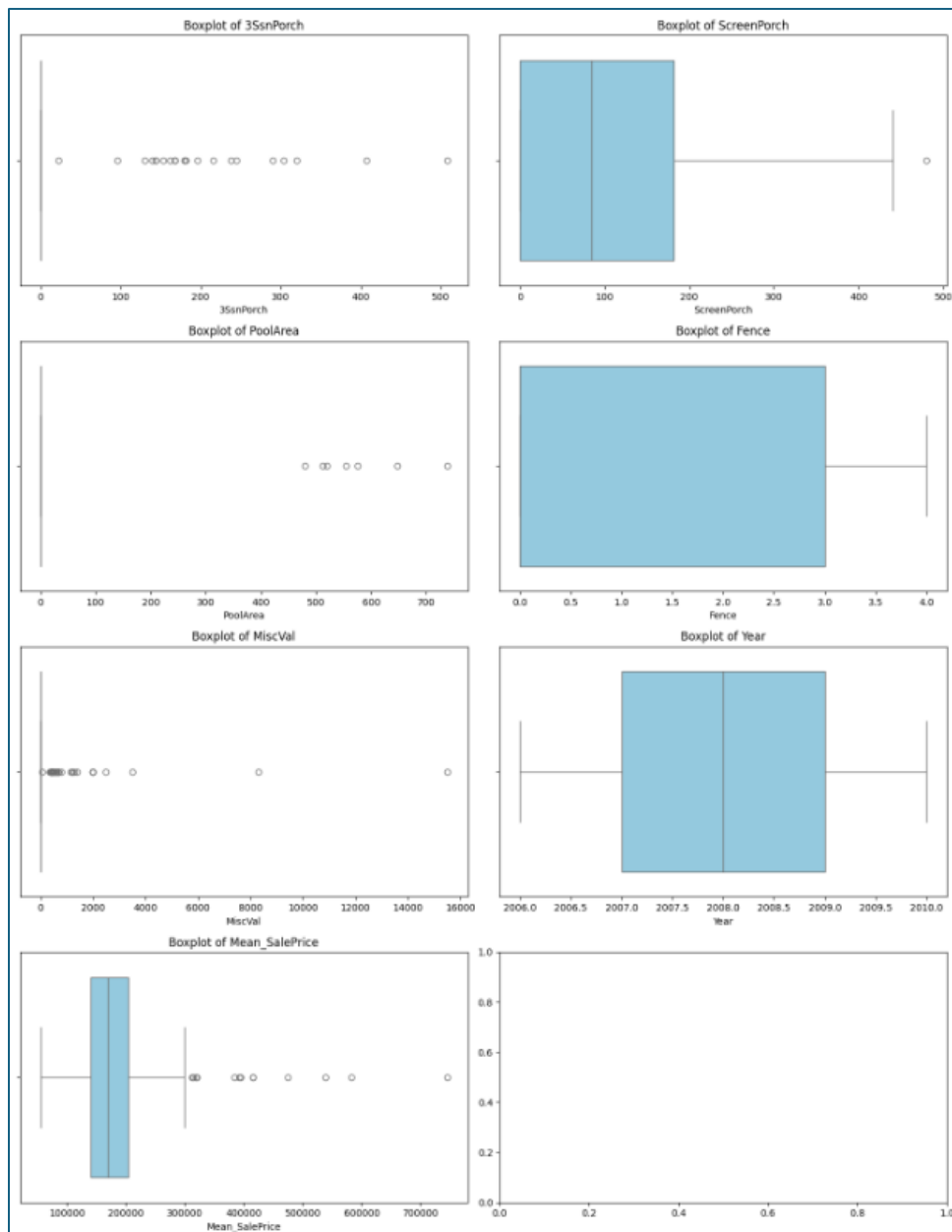
*Note.* This visualization presents the distribution of various property attributes in the dataset. It includes histograms with density plots for features such as 3-Season Porch (3SsnPorch), Screened Porch (ScreenPorch), Pool Area (PoolArea), Fence (Fence), Miscellaneous Value (MiscVal), and Year of Sale (Year).

**Figure 14**

*Correlation Map*



*Note.* This is a correlation heatmap that visualizes the relationships between different variables using a color-coded matrix such as dark red (1.00) indicates a perfect positive correlation, blue shades (negative values) indicate a negative correlation, light colors (near 0.00) indicate weak or no correlation.

**Figure 15***Outliers' detection*

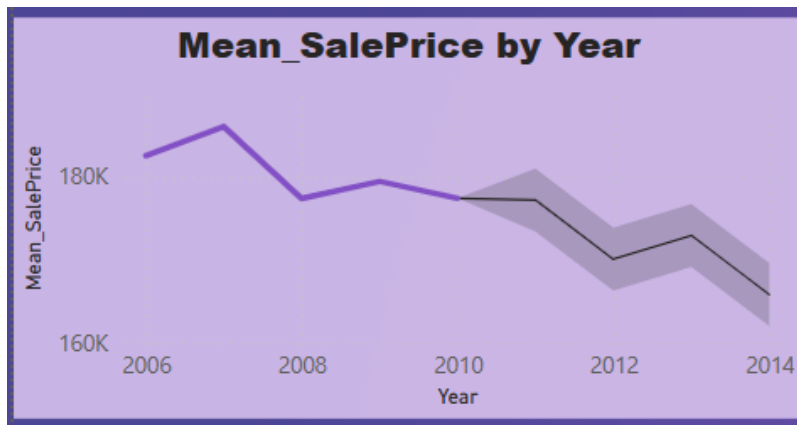
Note. This Graphs presents multiple boxplots used to visualize the distribution, central tendency, and presence of outliers for each feature. There are variables, such as PoolArea, MiscVal, and Mean\_SalePrice, have significant outliers.



A Real Estate Price Forecast has been created as tool that can be valuable for buyers, sellers, and investors looking to make informed decisions based on predicted price fluctuations.

**Figure 16**

*Real Estate Price Forecast*

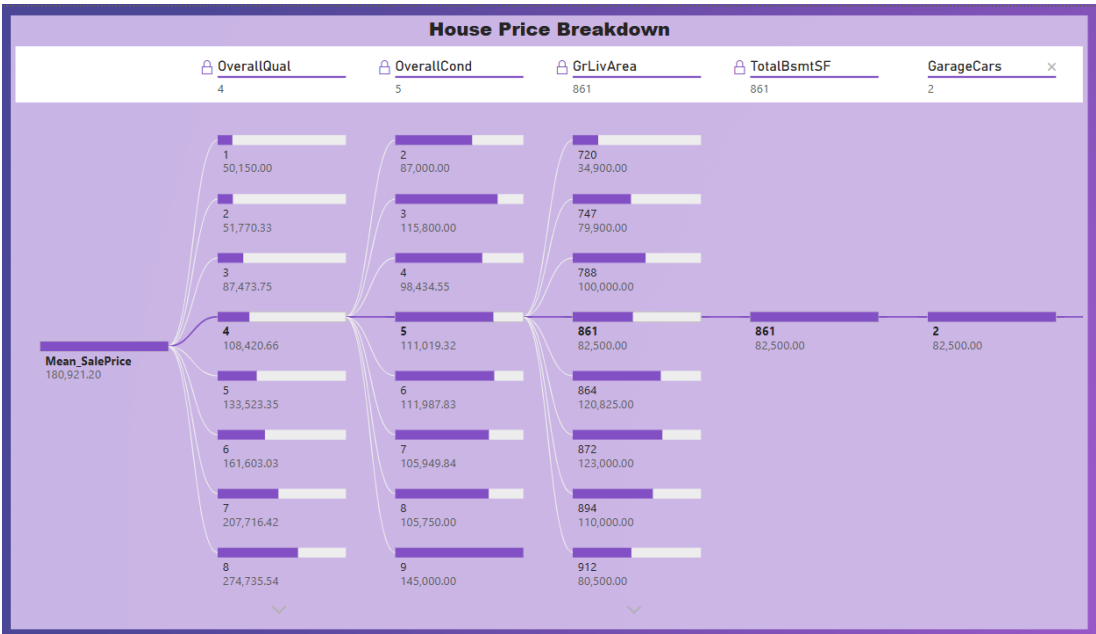


*Note.* This Line Chart presents a predictive analysis of the mean sale price of houses over time, utilizing historical data for real estate price estimation. The forecasted trend helps identify potential future price movements and trends in the housing market.

A decomposition Tree has been created to visualize as users navigate through different attribute levels, they can observe how each factor influences the average sale price, helping to identify patterns and trends in property valuation.

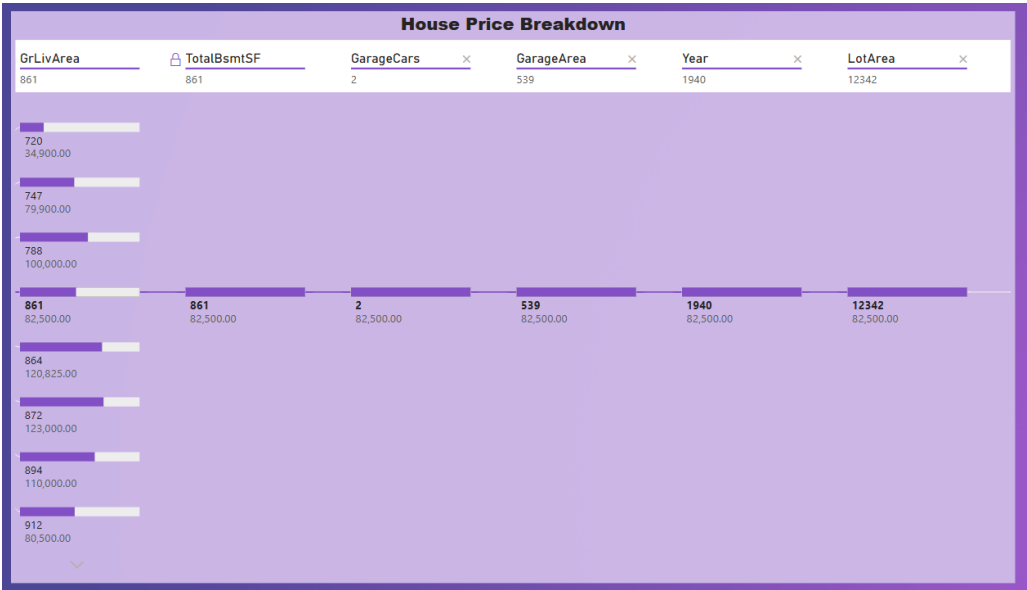
**Figure 17**

*Decomposition Tree House Price Breakdown.*



Note. This decomposition tree provides a detailed breakdown of the average sale price of houses based on various characteristics, such as Overall Quality (OverallQual), Overall Condition (OverallCond), Above-Ground Living Area (GrLivArea), Total Basement Area (TotalBsmtSF), and Number of Garage Cars (GarageCars).

**Figure 18**  
*Continue Decomposition Tree House Price Breakdown.*

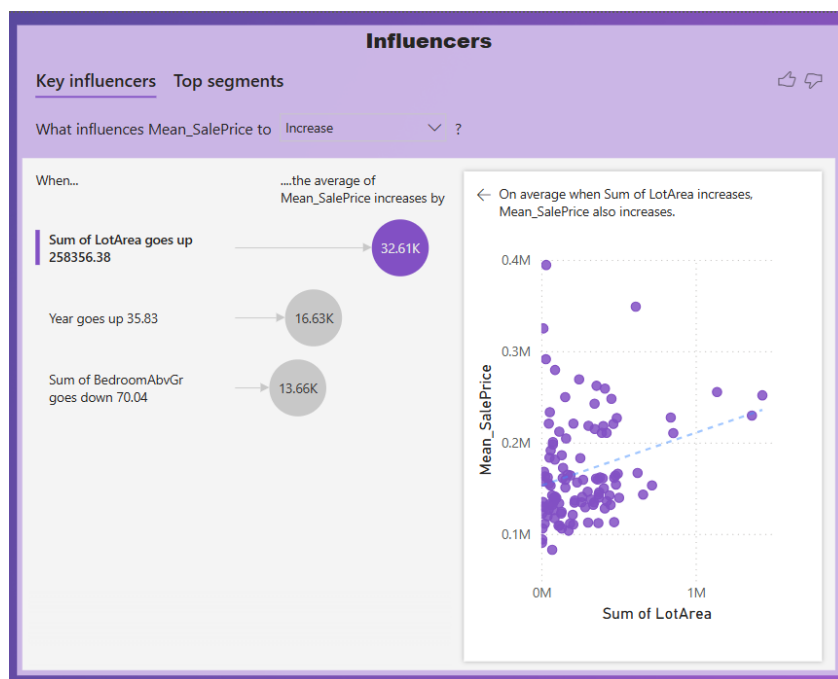


*Note.* This screen is continuation from the previous one, a detailed breakdown of the average sale price of houses based on various characteristics, such as (GarageCars), (GarageArea), (YearBuilt) and (LotArea).

### Key insights derived from the visualizations and advanced analysis.

**Figure 19**

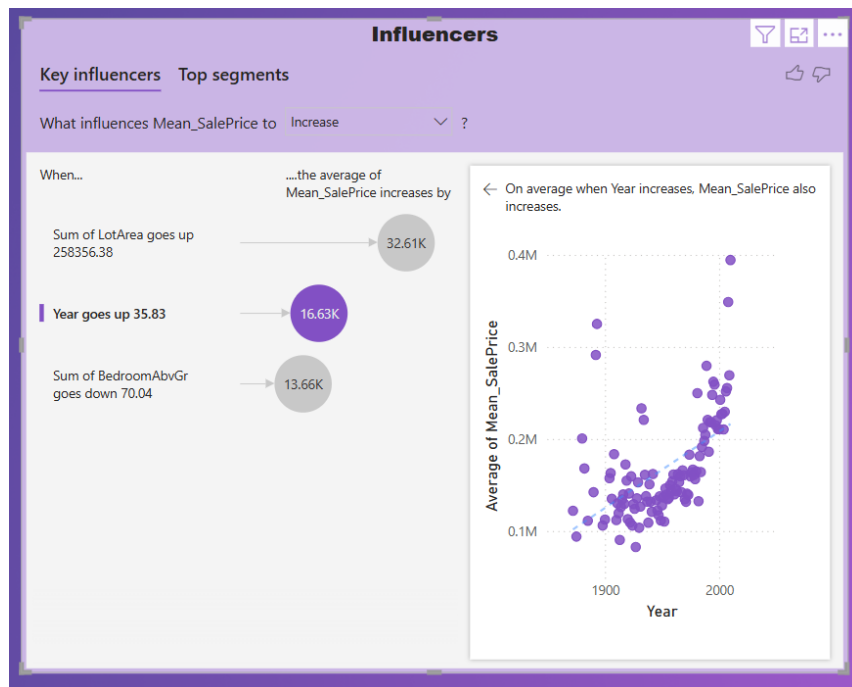
*Sum of Lot Area*



*Note.* This visualization shows how influence Mean\_SalePrices to increase when Sum of LotArea goes up 258356.38, the average of Mean\_SalePrices increase by 32.61K. This mean that on average when Sum of LotArea increases, Mean\_SalePrices also increases.

**Figure 20**

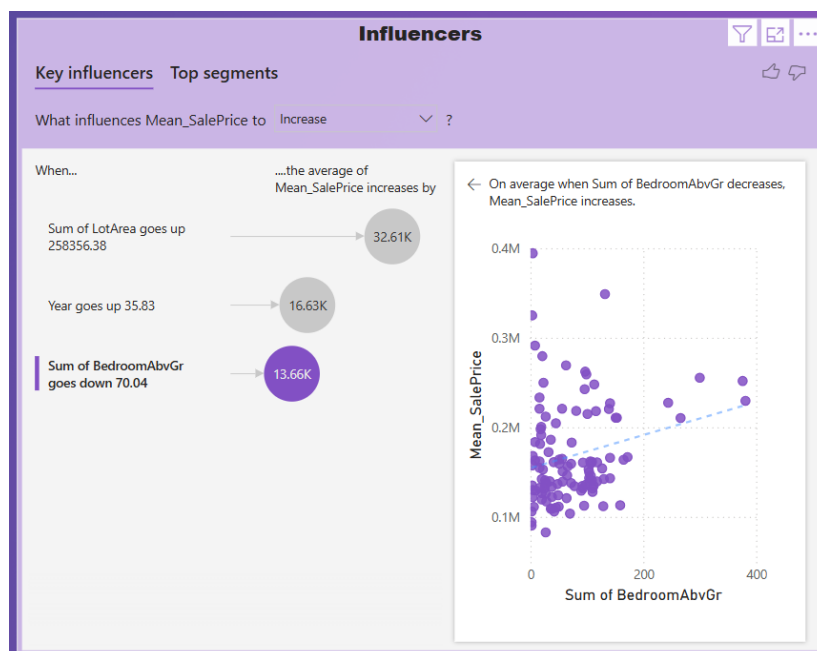
*Key influencer YearBuilt*



*Note.* This visualization shows how influence Mean\_SalePrices to increase when YearBuilt goes up 35.83, the average of Mean\_SalePrices increase by 16.63K. This mean that on average when YearBuilt increases, Mean\_SalePrices also increases.

**Figure 21**

*Influencer Sum of BedroomAvgr*



*Note.* This visualization shows how influence Mean\_SalePrices to increase when *Sum of BedroomAvgr* goes down 70.04, the average of Mean\_SalePrices increase by 13.66K. This mean that on average when *Sum of BedroomAvgr* decreases, Mean\_SalePrices increases.

### **Recommendations**

To make data-driven decisions, optimizing strategies for buying, selling, investing, and urban planning in the real estate market it can be useful Pricing Optimization for sellers and that way they can adjust pricing strategies based on key attributes that influence home values. Furthermore, Buyers can use the data to find homes that match their budget while maximizing property value.

Some exploration ideas are Geospatial Analysis, mapping price trends by neighborhood to identify high-growth areas, and incorporate economic indicators (interest rates, inflation) to see their impact on real estate prices.